

Boundary WZW evolution and the arithmetic Loewner source

A deterministic $SU(2)$ level-2 pilot and its realization limits

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23 September 2026

Abstract

We assemble two complementary calculations in the Wilson–Loewner program. For a normalized four-point boundary observable in the diagonal $SU(2)$ level-2 Wess–Zumino–Witten theory, the deterministic Loewner variation reduces to $\dot{M} = -\nu(2Q^2 + \dot{u}Q)M$, with $\nu = 1/4$ and Hermitian Q for real boundary spectators. We give the boundary-selected algebraic conformal block, the complete generator, composition and fusion, and a positive integrated balance for a constant driver. A moving driver can increase the same tensor norm. The Chern–Simons channel pairing is a different, transported metric, and the raw spatially smeared four-point kernel has a compact initial limit, so it does not supply the desired identity-initialized arithmetic transfer. Independently, the completed xi logarithmic derivative, translated into a known zero-free half-plane, defines normalized generalized Loewner flows and recovers the full arithmetic source, including prime-power delays. One flow is linearized by $\log \xi$. Removing the source translation is equivalent to the Riemann hypothesis (RH) at the level of whole-half-plane positivity; the arithmetic shift generator itself is not a positive-real Loewner driver, even under RH. We derive the available weighted operator bound and identify the missing cumulative boundary or canonical response with its unweighted norm.

Status. Synthesis of the 22 September calculations. The arithmetic realization remains open. Prepared with LLM assistance; Appendix A records provenance and 132 finite controls.

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1 The two calculations and the question they leave open

A physical realization of the shifted-zeta equation needs more than a geometric evolution with a positive pairing. It must specify a family of boundary operators, identify its initial operator and arithmetic generator, and relate its physical norm to the norm in which arithmetic contraction is required. This manuscript separates the results now available for the first two ingredients from that remaining operator question.

The physical calculation uses a fixed unitary boundary WZW theory. The boundary labels select a conformal block before any sign is tested, and the Ward and Knizhnik–Zamolodchikov (KZ) equations determine its response to a prescribed Loewner driver. This gives a completely explicit finite calculation, including the endpoint normalization. It does not construct the full sewing operator on the boundary affine modules.

The arithmetic calculation starts with the completed xi function and its logarithmic derivative. Positive-real function theory produces a genuine generalized Loewner evolution after a safe spectral translation. This construction contains the desired arithmetic coefficients exactly, but they enter through xi as input. An independently defined physical boundary response with this transfer and its norm remains to be found.

Object	Established here	Remaining obligation
Boundary WZW pilot	Selected block, complete deterministic generator, fusion and tensor norm balance	Full boundary-state sewing operator and its relation to a fixed input norm
Arithmetic source	Safe positive-real source, generalized Loewner flows, exact source recovery	Remove the safe translation without assuming the desired positivity
Arithmetic transfer	Exact shift equation and a weighted contraction bound	Cumulative response with the unweighted arithmetic norm

The standard boundary-CFT, KZ and Loewner results used below are credited to their sources. The explicit calculations are a synthesis of the project’s research notes, without a claim of literature novelty. Both results and counterexamples are scoped to the observables and identifications actually specified. In particular, failure of the raw four-point kernel does not exclude a construction retaining descendants or a different boundary readout.

This text is a separate companion to the parent Wilson–Loewner manuscript. It does not add sections to that manuscript. Section 8 states the present priority after both calculations: a cumulative response specified by the arithmetic source, with the WZW pilot retained as a physical benchmark.

2 A specified boundary theory and normalized slit observable

2.1 Boundary sectors and the chosen insertions

Use the diagonal unitary $SU(2)_2$ WZW theory, with central charge $c = 3/2$, maximally symmetric Cardy boundary conditions, and identity current gluing. The integrable labels are $0, 1/2, 1$. Writing V_j for the affine module including descendants, the boundary sectors are

$$\mathcal{H}_{ab} = \bigoplus_j N_{aj}^b V_j, \quad \mathcal{H}_{00} = V_0, \quad \mathcal{H}_{0,1/2} = V_{1/2}, \quad \mathcal{H}_{1/2,1/2} = V_0 \oplus V_1. \quad (2.1)$$

These are infinite-dimensional spaces. The boundary fusion input is the standard Cardy construction [1, 2]; it is distinct from the two-dimensional invariant tensor space used below.

Let D_t be the upper half-plane with a prescribed simple slit removed. Its hydrodynamically normalized map satisfies

$$\dot{g}_t(z) = \frac{2}{g_t(z) - u_t}, \quad g_0(z) = z, \quad u_0 = 0. \quad (2.2)$$

Assume u is C^1 and work on an interval where the slit, marked prime ends and ordered real spectators $0 < x < y$ remain well defined and unswallowed. The successive boundary labels in the uniformized half-plane are

$$0 \xrightarrow{u_t} \frac{1}{2} \xrightarrow{g_t(x)} 0 \xrightarrow{g_t(y)} \frac{1}{2} \xrightarrow{\infty} 0. \quad (2.3)$$

Each arrow carries the unique spin-1/2 boundary-changing primary in the corresponding sector. All four fields have weight $h = 3/16$. They are boundary fields, so there are no additional mirror insertions.

2.2 The normalization is part of the observable

Divide the four-field tensor amplitude by the two-field amplitude with the same tip and infinity fields, whose invariant contraction is normalized to one in the half-plane. Fix the remaining field normalization by the unit leading vacuum OPE coefficient used in Section 3. Set

$$a_t = g_t(x) - u_t, \quad b_t = g_t(y) - u_t, \quad J_x = g'_t(x), \quad J_y = g'_t(y). \quad (2.4)$$

If F is the normalized half-plane tensor amplitude, the slit observable is

$$M_t(x, y) = J_x^h J_y^h F(0, a_t, b_t, \infty). \quad (2.5)$$

Use the same uniformizing prime-end coordinate $g_t(z) - u_t$ and cutoff at the tip in numerator and denominator. Their tip-primary factors then cancel; no finite Euclidean derivative at the slit tip is invoked. The infinity insertion is $\lim_{z \rightarrow \infty} z^{2h} \phi(z)$, and its reciprocal-coordinate derivative is one by hydrodynamic normalization. Only the two spectator Jacobians remain. The common domain/Weyl anomaly factor cancels in the normalized ratio. There is no additional freely adjustable scalar function of t .

This uses the normalized slit-correlator framework of Alekseev–Bytsko–Izyurov [3]. Here the driver is deterministic; neither a Brownian driver nor an internal-group average, an Itô term, or a stochastic martingale theorem is imported.

2.3 What the Chern–Simons interpretation supplies

In holomorphic polarization on the doubled disk, the four marked fields give a sphere with four spin-1/2 punctures. The $SU(2)_2$ Chern–Simons conformal-block space has the two allowed intermediate channels 0, 1. A framed network preparing the first-pair vacuum channel selects the state used below. The relation between Wilson endpoints and conformal-block state spaces is established in the boundary formalism of Alekseev–Barmaz–Mnev [4].

Fix ribbon framings and local coordinates throughout. This supplies a Chern–Simons interpretation of the chiral data. It does not identify the two-dimensional slit with a bulk Wilson trajectory, or identify the finite channel space with either an affine boundary module or L^2 of an interval.

3 Explicit level-2 conformal blocks

Let $W = \text{Inv}((\mathbb{C}^2)^{\otimes 4})$ have the orthonormal basis

$$e_0 = s_{01}s_{23}, \quad e_1 = \frac{t_{01}^+ t_{23}^- + t_{01}^- t_{23}^+ - t_{01}^0 t_{23}^0}{\sqrt{3}}, \quad (3.1)$$

where $s = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ and t^+, t^0, t^- are the normalized triplets. Use $t^a = \sigma^a/\sqrt{2}$ and $T_{ij} = \sum_a t_i^a t_j^a$. Put $\nu = (k+2)^{-1} = 1/4$. Then $C_{1/2} = 3/2$ and $h = \nu C_{1/2}/2 = 3/16$. In this basis,

$$D = T_{01} = \begin{pmatrix} -3/2 & 0 \\ 0 & 1/2 \end{pmatrix}, \quad C = T_{02} = \begin{pmatrix} 0 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1 \end{pmatrix}, \quad B = T_{12} = \begin{pmatrix} 0 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1 \end{pmatrix}. \quad (3.2)$$

In particular $D + C + B = -3I/2$. Replacing these generators by $\sigma^a/2$ without changing the KZ coefficient would incorrectly halve the vacuum OPE exponent.

Translation and scaling covariance give

$$F(0, a, b, \infty) = b^{-2h} f(r), \quad r = a/b, \quad f'(r) = \nu \left(\frac{D}{r} + \frac{B}{r-1} \right) f(r). \quad (3.3)$$

The first adjacent pair in (2.3) has label 0 on both outside intervals. Its OPE lies in $\mathcal{H}_{00} = V_0$ and therefore selects

$$f^{(0)}(r) = r^{-3/8} (e_0 + O(r)), \quad (3.4)$$

with no independent spin-one Frobenius branch. This selection is fixed by the boundary labels, rather than by choosing initial data for favorable signs. A single selected block generally has both tensor components.

For $0 < r < 1$, write $s = \sqrt{1-r}$ and $q = [r(1-r)]^{-3/8}$, using positive real roots. The two unit-leading Frobenius solutions are

$$f^{(0)}(r) = \frac{q}{2\sqrt{2}} \begin{pmatrix} (1+s)^{3/2} \\ -\sqrt{3}(1-s)\sqrt{1+s} \end{pmatrix}, \quad (3.5)$$

$$f^{(1)}(r) = \frac{q}{\sqrt{6}} \begin{pmatrix} -(1-s)^{3/2} \\ \sqrt{3}(1+s)\sqrt{1-s} \end{pmatrix}. \quad (3.6)$$

The second solution has leading behavior $r^{1/8} e_1$. Both columns are useful for transport and fusion, although the specified boundary conditions select (3.5).

For a direct derivation, eliminate the second component of (3.3) and write the first as $qP(r)$. The scalar equation becomes

$$r(1-r)P'' - \frac{1}{2}P' - \frac{3}{16}P = 0. \quad (3.7)$$

After $s = \sqrt{1-r}$ it is $(1-s^2)P_{ss} + sP_s - 3P/4 = 0$, solved by $(1+s)^{3/2}$ and $(1-s)^{3/2}$. Substitution in the first KZ component and normalization of the leading tensors give (3.5)–(3.6).

An independent local construction writes $f = r^\lambda \sum_{n \geq 0} v_n r^n$ and uses

$$[(\lambda + n)I - \nu D]v_n = -\nu B \sum_{m=0}^{n-1} v_m, \quad n \geq 1. \quad (3.8)$$

Take $(\lambda, v_0) = (-3/8, e_0)$ or $(1/8, e_1)$. The exponent difference is $1/2$, so the displayed inverses exist. For example,

$$f^{(0)}(r) = r^{-3/8} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + r \begin{pmatrix} 0 \\ -\sqrt{3}/4 \end{pmatrix} + r^2 \begin{pmatrix} 3/64 \\ -\sqrt{3}/8 \end{pmatrix} + O(r^3) \right]. \quad (3.9)$$

At $(u, x, y) = (0, 1, 2)$, the physical selected state is

$$M_0 = 2^{-3/8} f^{(0)}(1/2) \simeq (1.02266239412001, -0.30390758736355)^T. \quad (3.10)$$

4 Complete deterministic evolution and the tensor norm

The relative coordinates and Jacobians satisfy

$$\dot{a} = 2/a - \dot{u}, \quad \dot{b} = 2/b - \dot{u}, \quad \partial_t \log J_x = -2/a^2, \quad \partial_t \log J_y = -2/b^2. \quad (4.1)$$

The two finite-coordinate KZ connections are

$$A_a = \nu \left(\frac{D}{a} + \frac{B}{a-b} \right), \quad A_b = \nu \left(\frac{C}{b} + \frac{B}{b-a} \right). \quad (4.2)$$

Translation invariance includes the moving tip: $\partial_u F = -(\partial_a + \partial_b)F = -\nu(D/a + C/b)F$. Differentiating (2.5) therefore yields $\dot{M} = GM$, with

$$G = \nu \left[\left(\frac{2}{a^2} - \frac{\dot{u}}{a} \right) D + \left(\frac{2}{b^2} - \frac{\dot{u}}{b} \right) C - \frac{2B}{ab} \right] - 2h \left(\frac{1}{a^2} + \frac{1}{b^2} \right) I. \quad (4.3)$$

The B coefficient follows by subtracting the two spectator velocities; their common $-\dot{u}$ cancels in that coefficient, but not in the D, C terms. The final scalar comes from the primary Jacobians.

Proposition 4.1 (Deterministic square reduction). *On every compact interval before collision or swallowing, the normalized observable satisfies*

$$\boxed{\dot{M}_t = -\nu(2Q_t^2 + \dot{u}_t Q_t)M_t, \quad Q_t = \frac{D}{a_t} + \frac{C}{b_t}, \quad \nu = \frac{1}{4}.} \quad (4.4)$$

The associated two-dimensional initial-value problem has a unique invertible propagator.

Proof. The matrices in (3.2) obey

$$D^2 = \frac{3}{4}I - D, \quad C^2 = \frac{3}{4}I - C, \quad DC + CD = B. \quad (4.5)$$

Expanding $-2\nu(D/a + C/b)^2$ reproduces the terms in (4.3) independent of $\dot{u}$, since $2h = 3\nu/2$. The remaining term is $-\nu\dot{u}Q$. The coefficients are continuous and bounded on the stated compact time interval, giving the usual matrix-ODE existence and uniqueness. $\square$

The same algebraic reduction extends to a spin-1/2 tip with finitely many real boundary primary spectators, using $Q = \sum_i T_{0i}/w_i$, $h_i = \nu C_i/2$, $T_{0i}^2 = C_i I/2 - T_{0i}$ and $\{T_{0i}, T_{0j}\} = T_{ij}$ for $i \neq j$. Boundary conditions and integrable channels still require specification. Complex bulk or mirror coordinates need not give Hermitian Q .

4.1 An exact balance for a constant driver

For the real boundary coordinates, $Q = Q^\dagger$ in the fixed tensor metric of W . If $\dot{u} = 0$, differentiation of the norm and integration give

$$\boxed{\|M_T\|_W^2 + 4\nu \int_0^T \|Q_t M_t\|_W^2 dt = \|M_0\|_W^2.} \quad (4.6)$$

This holds for every initial evaluated vector in W , without assuming that matrices at different times commute. For $a = 1, b = 2$,

$$G_{\dot{u}=0} = \begin{pmatrix} -39/32 & -3\sqrt{3}/16 \\ -3\sqrt{3}/16 & -3/32 \end{pmatrix}. \quad (4.7)$$

More generally $\det Q = -3(1/a - 1/b)^2/4$, so distinct spectators give an invertible Q . The constant-driver evolution is nontrivial.

4.2 Moving drivers can increase the same norm

The general identity is

$$\frac{d}{dt} \|M\|_W^2 = -4\nu \|QM\|_W^2 - 2\nu \dot{u} \langle M, QM \rangle_W. \quad (4.8)$$

At $a = 1, b = 2, \dot{u} = 4$,

$$G = \begin{pmatrix} 9/32 & \sqrt{3}/16 \\ \sqrt{3}/16 & -3/32 \end{pmatrix}. \quad (4.9)$$

For the selected state (3.10), the initial derivative of its squared norm is approximately $0.436388495438834 > 0$. This disproves an all-driver contraction claim even for the chosen Cardy preparation. It does not imply monotone growth at later times.

Completing the square gives the general bound

$$G = -2\nu(Q + \dot{u}I/4)^2 + \nu\dot{u}^2 I/8, \quad \|U(T, 0)\|_W \leq \exp\left(\frac{\nu}{8} \int_0^T \dot{u}^2 dt\right). \quad (4.10)$$

Multiplying M by the inverse scalar on the right would make this pairing contractive for every C^1 driver. That factor is additional to (2.5); no physical partition function has been shown to supply it. It therefore cannot stand in for a fixed arithmetic normalization.

5 Composition, gluing and the spatial-kernel limitation

5.1 Transport and fusion

Write $\Phi(r) = [f^{(0)}(r), f^{(1)}(r)]$ and

$$E_t = J_x^h J_y^h b_t^{-2h} \Phi(a_t/b_t). \quad (5.1)$$

In a single original-coordinate frame, the exact propagator is

$$U(t, s) = E_t E_s^{-1}, \quad U(t, s)U(s, r) = U(t, r), \quad U(s, s) = I. \quad (5.2)$$

It is also the time-ordered exponential of the generator. Restarting the Loewner map requires the translated remaining driver and the evolved marked points, rather than a reset to the original geometry. The primary-Jacobian chain rule gives the same composition.

For fusion, define

$$R = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{pmatrix}, \quad Z = \text{diag}(1, -1), \quad \Phi_{12}(r) = RZ\Phi(1-r)Z. \quad (5.3)$$

Its columns have unit leading tensors in the 12 channels. Direct substitution of (3.5)–(3.6) yields

$$\Phi(r) = \Phi_{12}(r)\mathbf{C}, \quad \mathbf{C} = \begin{pmatrix} 1/\sqrt{2} & -\sqrt{2/3} \\ \sqrt{3/8} & 1/\sqrt{2} \end{pmatrix}. \quad (5.4)$$

With $N = \text{diag}(1, \sqrt{3}/2)$, the unitary channel gauge gives

$$N^{-1}\mathbf{C}N = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}. \quad (5.5)$$

The signs depend on the chosen tensor basis and channel gauge.

5.2 The channel metric is not the fixed tensor metric

In raw Frobenius coefficients c, d , the positive unitary Chern–Simons pairing, normalized by vacuum norm one, is

$$\langle c, d \rangle_{\text{CS}} = c^\dagger H_c d, \quad H_c = \text{diag}(1, 4/3). \quad (5.6)$$

Local monodromies have the distinct phases of the exponents $-3/8, 1/8$, forcing the invariant metric to be diagonal. Requiring fusion to preserve it fixes the ratio $4/3$. Equivalently, (5.5) is unitary on $N^{-1}c$. Orientation reversal conjugates the amplitude and sewing contracts the matching channel indices with this pairing. The selected vacuum has norm one and equal-magnitude coefficients $1/\sqrt{2}$ in the normalized crossed channel.

For evaluated tensors $M = E_t c$, the same pairing becomes

$$H_t = E_t^{-\dagger} H_c E_t^{-1}, \quad U(t, s)^\dagger H_t U(t, s) = H_s. \quad (5.7)$$

Thus the CS channel norm is preserved while the evaluated tensor can decrease in the fixed W norm of (4.6). Neither metric has been identified with the arithmetic L^2 norm. Likewise a scalar polarization $v^\dagger M$ with fixed $v \in W$ is different from a channel amplitude $d^\dagger H_c c$.

5.3 Spatial superposition does not create the initial identity

For disjoint compact boundary intervals $I_x < I_y$ away from the hull,

$$M_t[f, g] = \int_{I_x} \int_{I_y} f(x)g(y)M_t(x, y) \, dy \, dx \quad (5.8)$$

is a well-defined superposition for smooth compactly supported tests. Its derivative is the corresponding integral of $G_t(x, y)M_t(x, y)$. Since G_t depends on the positions, this single smeared two-vector does not satisfy a closed 2×2 equation. Retaining all positions gives a multiplication evolution, and integrating (4.6) against a positive spatial measure gives a direct-integral norm balance. Arbitrary functions in that enlarged space have not been prepared as physical boundary states.

There is a stronger initial-condition test. Fix a compact interval $I \subset (0, \infty)$ away from the hull and a polarization $v \in W$. Set $\mathcal{K}_t(x, y) = v^\dagger M_t(x, y)$ for $x < y$, and zero for the opposite order. This chosen ordering does not derive arithmetic causality.

Proposition 5.1 (The raw ordered kernel has a compact initial limit). *On a compact time interval where g_t is regular on I , the integral operator with kernel $\mathcal{K}_t$ is Hilbert–Schmidt. As $t \downarrow 0$ it converges in Hilbert–Schmidt norm to its explicit compact operator at $t = 0$, and therefore cannot have the identity as its initial limit on the infinite-dimensional space $L^2(I)$.*

Proof. The worst diagonal singularity is the 12 identity OPE, of exponent $-2h = -3/8$. Uniform regularity of the coordinate map and spectator Jacobians gives

$$|\mathcal{K}_t(x, y)| \leq C_T(1 + |x - y|^{-3/8}). \quad (5.9)$$

The square has exponent $-3/4$, integrable on $I \times I$. Pointwise continuity off the diagonal and dominated convergence yield the stated Hilbert–Schmidt limit. That limit is compact and cannot equal the identity. $\square$

A finite scalar normalization or a regular change of time cannot alter this explicit limit. The result does not exclude a new sewing construction with descendants, a separately derived identity term, or a distinct singular limiting regime. Compact operators can converge strongly to the identity in other settings; the present obstruction also uses convergence to the specified compact initial operator.

6 The arithmetic source defines generalized Loewner flows

6.1 The full source of the shift equation

Define

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad H(p) = \xi\left(\frac{1}{2} + p\right), \quad m(p) = \frac{H'(p)}{H(p)}. \quad (6.1)$$

The target transfer and its logarithmic generator are

$$K_\omega(p) = \frac{H(p-\omega)}{H(p+\omega)}, \quad a_\omega(p) = m(p-\omega) + m(p+\omega), \quad \partial_\omega K_\omega = -a_\omega K_\omega, \quad (6.2)$$

with $K_0 = 1$. The meromorphic identity is first used on a zero-free right half-plane. Individual poles of a_ω can cancel in the product with K_ω .

The completed logarithmic derivative is

$$m(p) = \frac{1}{p+1/2} + \frac{1}{p-1/2} - \frac{1}{2}\log \pi + \frac{1}{2}\psi\left(\frac{1}{4} + \frac{p}{2}\right) + \frac{\zeta'}{\zeta}\left(\frac{1}{2} + p\right), \quad (6.3)$$

where ψ is the digamma function. Apparent singularities at removable points of the completed function must be treated together. In the region $\operatorname{Re} p > 1/2 + \omega$, the prime part of a_ω is

$$-2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \cosh(\omega \log n) e^{-p \log n}. \quad (6.4)$$

The inverse Laplace source therefore contains the delays $\log n$ with their full coefficients, in addition to the gamma, rational and local terms. It is the logarithmic source that has the prime-power comb; the transfer itself has the fuller multiplicative integer comb.

6.2 A safe positive-real function, and its limiting condition

Write the xi zeros, with multiplicity, as $\rho = 1/2 + \alpha_\rho + i\gamma_\rho$. The centered product gives

$$\operatorname{Re} m(x + iy) = \sum_\rho \frac{x - \alpha_\rho}{(x - \alpha_\rho)^2 + (y - \gamma_\rho)^2}. \quad (6.5)$$

The complex series uses symmetric grouping; the real-part series is absolutely convergent. Functional symmetry removes an additional real constant. No restriction of the sum to assumed critical zeros is made.

Let $\mathbb{H}_R = \{p : \operatorname{Re} p > 0\}$ and $m_\eta(p) = m(p + \eta)$ for $\eta \geq 0$. If ξ has no zero with $\operatorname{Re} \rho > 1/2 + \eta$, every numerator in (6.5), evaluated at $p + \eta$, is positive on $\mathbb{H}_R$. Conversely such a zero would be an interior pole of m_η with a positive integer residue. Consequently

$$\begin{aligned} m_\eta \text{ is holomorphic with } \operatorname{Re} m_\eta > 0 \text{ on } \mathbb{H}_R \\ \iff \xi \text{ has no zero with } \operatorname{Re} \rho > \frac{1}{2} + \eta. \end{aligned} \quad (6.6)$$

For $\eta \geq 1/2$ this is unconditional. At $\eta = 0$ it is equivalent to RH, using functional symmetry. This is the classical positivity criterion of Lagarias [5, 6], applied in centered coordinates; it is not a new RH criterion.

6.3 A normalized radial evolution and its driving measure

Use the Cayley maps and normalization

$$C_{\mathbb{D}}(w) = \frac{1+w}{1-w}, \quad \chi(p) = \frac{p-1}{p+1}, \quad c_\eta = m(1+\eta), \quad P_\eta(w) = \frac{m(\eta + C_{\mathbb{D}}(w))}{c_\eta}. \quad (6.7)$$

Here $c_\eta > 0$ is unconditional for $\eta \geq 0$. If $\eta \geq 1/2$, then P_η is holomorphic on the disk, has positive real part, and satisfies $P_\eta(0) = 1$. The classical generator criterion [7] therefore gives the global univalent disk self-map semigroup

$$\partial_\tau \varphi_\tau(w) = -\varphi_\tau(w) P_\eta(\varphi_\tau(w)), \quad \varphi_0(w) = w, \quad \varphi'_\tau(0) = e^{-\tau}. \quad (6.8)$$

This is the inward evolution-family convention for generalized radial Loewner evolution, not a claim of a single-slit chordal driver.

The Herglotz probability measure is fixed by

$$P_\eta(w) = \int_{|z|=1} \frac{z+w}{z-w} d\mu_\eta(z). \quad (6.9)$$

On a safe line it has density

$$d\mu_\eta(e^{i\theta}) = \frac{\operatorname{Re} m(\eta + i \cot(\theta/2))}{2\pi c_\eta} d\theta. \quad (6.10)$$

The possible atom at 1 vanishes since $m(\eta + x)/x \rightarrow 0$ as $x \rightarrow +\infty$. At all other boundary points the source continues analytically, excluding another singular part.

Under RH the zero-shift measure has the atomic form

$$z_\gamma = \frac{i\gamma - 1}{i\gamma + 1}, \quad \mu_0 = \sum_{\gamma \text{ of either sign}} \frac{1}{c_0(1 + \gamma^2)} \delta_{z_\gamma}. \quad (6.11)$$

The sum includes multiplicities. Pairing the two signs gives $m(p) = \sum_{\gamma > 0} 2p/(p^2 + \gamma^2)$, and evaluation at $p = 1$ shows total mass one. The summable weights matter: a real-line measure with unit mass at every zero would have infinite total mass.

Most directly, on $\operatorname{Re} p > \eta + \omega$ the complete arithmetic source is

$$a_\omega(p) = c_\eta [P_\eta(\chi(p - \omega - \eta)) + P_\eta(\chi(p + \omega - \eta))]. \quad (6.12)$$

Thus the genuine Loewner source recovers every term of (6.3)–(6.4). Its arithmetic content is prescribed by ξ , rather than derived from independent field data.

6.4 A flow linearized by the logarithm of xi

Fix $\eta \geq 1/2$ and choose the analytic branch $L_\eta(p) = \log H(p + \eta)$ real on the positive real axis. It is univalent on $\mathbb{H}_R$, because for $p \neq q$,

$$\frac{L_\eta(p) - L_\eta(q)}{p - q} = \int_0^1 m(q + s(p - q) + \eta) ds \quad (6.13)$$

has positive real part. The reciprocal source defines another semigroup:

$$\partial_\tau \psi_\tau(p) = \frac{c_\eta}{m(\psi_\tau(p) + \eta)}, \quad \psi_0(p) = p. \quad (6.14)$$

Under the Cayley map its generator is $(1 - w)^2/[2P_\eta(w)]$. Since $\text{Re}(1/P_\eta) > 0$, this is a Berkson–Porta generator with boundary fixed point 1, giving global self-maps of $\mathbb{H}_R$. Direct differentiation gives

$$\boxed{L_\eta(\psi_\tau(p)) = L_\eta(p) + c_\eta \tau, \quad H(\psi_\tau(p) + \eta) = e^{c_\eta \tau} H(p + \eta).} \quad (6.15)$$

The arithmetic transfer can also be written, in the domain of (6.12), as

$$K_\omega(p) = \exp [L_\eta(p - \omega - \eta) - L_\eta(p + \omega - \eta)]. \quad (6.16)$$

This identifies the common analytic potential. It does not identify arithmetic shift with geometric time, or spectral position with physical boundary position.

7 Continuation and the arithmetic operator norm

7.1 Four parameters with different meanings

The boundary capacity t in (2.2), the generalized Loewner time τ in (6.8) or (6.14), the source translation η , and the arithmetic shift ω are distinct. No equality between them, or between them and the spatial window length, has been derived.

Differentiating the normalized source gives

$$\partial_\eta P_\eta(w) = \frac{(1 - w)^2}{2} \partial_w P_\eta(w) - \kappa_\eta P_\eta(w), \quad \kappa_\eta = \frac{m'(1 + \eta)}{m(1 + \eta)}. \quad (7.1)$$

For $h \geq 0$, it integrates exactly to

$$P_{\eta+h}(\chi(p)) = \frac{c_\eta}{c_{\eta+h}} P_\eta(\chi(p + h)). \quad (7.2)$$

Right translation preserves the half-plane and the positive-real property. On safe lines, $u_\eta(y) = \text{Re } m(\eta + iy)$ obeys Poisson smoothing,

$$u_{\eta+h}(y) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{h}{h^2 + (y - v)^2} u_\eta(v) dv, \quad h > 0. \quad (7.3)$$

There is no extra linear harmonic term at infinity, by the xi asymptotic. Reversing this smoothing need not preserve positivity.

For a concrete control, the even real entire polynomial

$$H_\delta(p) = ((p - \delta)^2 + \gamma^2)((p + \delta)^2 + \gamma^2) \quad (7.4)$$

has a positive-real shifted logarithmic derivative when $\eta > \delta$, but interior poles when $\eta < \delta$. At $\delta = 1/4, \gamma = 2$, the shift $1/2$ is safe while at $\eta = 1/10$ a disk pole has modulus approximately 0.941951. This excludes a general continuation argument based only on functional symmetries and positivity at a safe shift; it is not a claim about the actual xi zeros. Even $H(p) = p^2 + \gamma^2$ has $m'(x) = 2(\gamma^2 - x^2)/(x^2 + \gamma^2)^2$ of both signs, so simple pointwise monotonicity cannot replace the missing argument.

7.2 Endpoint paths and operator actions do not yet match

The endpoints used in (6.12) satisfy

$$w_{\pm}(\omega) = \chi(p \pm \omega - \eta), \quad \partial_{\omega} w_{\pm} = \pm \frac{1}{2}(1 - w_{\pm})^2. \quad (7.5)$$

These are opposite half-plane translations. At $w = 0$ their velocities are $\pm 1/2$, while the radial field $-wP_{\eta}(w)$ vanishes. A scalar change of clock cannot identify these vector fields on the whole domain. The reciprocal-source flow has a position-dependent velocity instead of either fixed translation.

Likewise, the arithmetic action in Laplace coordinates is multiplication, $F \mapsto K_{\omega}F$, whereas a self-map acts by composition. If $W(p)F(\psi(p)) = K_{\omega}(p)F(p)$ for every analytic F , testing $F = 1, p$ forces $W = K_{\omega}$ and $\psi(p) = p$ where $K_{\omega} \neq 0$. This algebraic test does not presume that $1, p$ are half-plane Hardy vectors. It excludes the direct identification on all analytic functions, not a further boundary projection, integral transform or intertwiner.

7.3 Cumulative positivity is the appropriate target

Even under RH, $m(p - \omega)$ has poles at $p = \omega + i\gamma$ inside $\mathbb{H}_R$ for every $\omega > 0$. The other source term does not cancel them. Thus a_{ω} is not a holomorphic positive-real driver on that half-plane. Zeros of K_{ω} cancel those poles in (6.2).

The failure is also visible away from the poles. For $H(p) = p^2 + 4$, K_{ω} is a finite half-plane Blaschke product, but at $p = 1/4 + 2i$, $\omega = 1/2$, exact arithmetic gives

$$|K_{\omega}(p)|^2 = \frac{257}{2385} < 1, \quad \operatorname{Re} a_{\omega}(p) = -\frac{538768}{204315} < 0. \quad (7.6)$$

Hence the squared modulus is locally increasing in ω despite remaining below one. Cumulative contraction does not require contractive multiplicative increments.

The cumulative positive-real candidate is the Cayley transform

$$Z_{\omega} = \frac{1 - K_{\omega}}{1 + K_{\omega}}, \quad \operatorname{Re} Z_{\omega} = \frac{1 - |K_{\omega}|^2}{|1 + K_{\omega}|^2}, \quad \partial_{\omega} Z_{\omega} = \frac{a_{\omega}}{2}(1 - Z_{\omega}^2). \quad (7.7)$$

The Riccati identity holds on regular regions and through removable singularities. It is not an independent proof of positivity.

On a finite causal window, the corresponding target is $V_{0,L} = I$, $\partial_{\omega} V_{\omega,L} = -G_{\omega,L}V_{\omega,L}$. Its accumulated defect obeys, on admissible inputs,

$$I - V_{\omega,L}^* V_{\omega,L} = \int_0^{\omega} V_{s,L}^* (G_{s,L}^* + G_{s,L}) V_{s,L} ds. \quad (7.8)$$

Generator pairings are understood as forms when separate operator domains are unavailable. The initial limit is strong, with differentiation on smooth tests; no operator-norm derivative at zero is assumed. A physical construction must account for the full accumulated quantity, not impose the false pointwise condition excluded by (7.6).

7.4 The unconditional weighted operator control

Fix $0 \leq \omega \leq \Omega$ and translate the transfer by $b \geq \Omega + 1/2$. Both terms in $a_\omega(p + b)$ are then positive-real on $\mathbb{H}_R$. Integrating (6.2) from $K_0 = 1$ gives

$$|K_\omega(p + b)| \leq 1, \quad \operatorname{Re} p > 0. \quad (7.9)$$

It is an analytic contractive multiplier on the causal Hardy/Laplace space. On $L^2(0, L)$, define $(E_b f)(x) = e^{bx} f(x)$. The Laplace translation rule and causal compression give

$$V_{\omega,b,L} = E_{-b} V_{\omega,L} E_b, \quad \|V_{\omega,b,L}\| \leq 1, \quad \boxed{\|V_{\omega,L}\| \leq e^{bL}}. \quad (7.10)$$

This also defines the bounded unshifted finite-window operator consistently with its causal inverse Laplace kernel. Pointwise multiplier convergence and dominated convergence give the strong identity limit at zero shift.

The last estimate is an upper bound, not a claim of actual exponential growth. Its length-dependent cost is precisely why safe-line positivity does not give the unweighted contraction. The unconditional b remains at least $1/2$, so the bound does not tend to one as $L \rightarrow \infty$.

8 Current research priority and decision criteria

The finite WZW pilot is complete: it has a boundary-selected initial block, a fully normalized observable, closed deterministic evolution, explicit fusion and composition, and a spatial-smearing test. Its constant-driver balance is in a specified tensor pairing. Its proposed extension is a slit sewing map on the boundary affine modules, retaining descendants and endpoint fields, with a physical reflected pairing. Such a construction must reproduce the primary matrix elements above and explain its relation to both metrics in Section 5.

For arithmetic matching, the direct source calculation now gives a sharper starting point than further sewing without an arithmetic specification. The next bounded task is a cumulative boundary or canonical response for K_ω or Z_ω , starting from the complete source on a safe line. Its deliverable is an operator dictionary in a fixed Hilbert space and an explicit norm balance, including the cost of removing the weight in (7.10). Three tests make that task concrete:

1. Recover the identity initial operator and the full local, gamma and rational terms on $0 < L < \log 2$, before any prime delay is visible.
2. On crossing $L = \log 2$, recover the exact first-delay coefficient $-\sqrt{2} \log 2 \cosh(\omega \log 2)$ in the logarithmic generator, with control of every boundary or storage term.
3. Produce a cumulative positive balance compatible with (7.6), rather than requiring positive instantaneous multiplicative increments.

Canonical-system work already provides a related comparison. Suzuki [8] studies $\Theta_\omega(z) = K_\omega(-iz)$, its meromorphic innerness and the associated canonical systems. His explicit unconditional construction has $\omega > 1$. This is different from unconditional innerness for $\omega \geq 1/2$, and from the safe source translation $\eta \geq 1/2$. Innerness for every positive arithmetic shift is RH-equivalent. Canonical depth must therefore not be silently identified with ω or τ .

The modular-surface scattering realization at $\omega = 1/2$, already recorded in the parent investigation, remains an endpoint comparison. It does not supply a continuous family at smaller shifts.

Likewise, postulating the measure (6.11) without RH would assume the desired positive-real property instead of deriving it.

An independent realization must specify its operator and positive norm without assuming the contractivity it is meant to explain. The present work provides an explicit physical benchmark, an exact arithmetic source for comparison, and several rejection tests for proposed identifications. It does not establish a matching physical transfer, its unweighted cumulative norm identity, or an all-depth continuation.

A Recorded controls, provenance and acknowledgements

A.1 What the numerical records establish

Two separate programs accompany the underlying 22 September research notes. Their unchanged records contain 132 passing cases: 11 exact rational controls and 121 floating diagnostics. These counts do not belong to the parent manuscript's original 299-case suite, and this editorial synthesis does not claim a new numerical experiment.

The WZW program has 72 cases, seven rational and 65 floating. It checks the independent 16-dimensional Pauli construction, algebraic blocks against the Frobenius recurrence, KZ derivatives, fusion and both pairings, the square reduction, and full two-column deterministic evolution. Five drivers are integrated at two resolutions, with composition and norm controls. Negative controls detect missing tip drift, omitted Jacobians, scalar component closure, and an unrestricted contraction claim.

For $x = 1, y = 2, T = 0.2$, the recorded values are shown below. The initial selected tensor norm squared is 1.1381981940 in every row.

Driver	Final selected norm squared	Largest singular value of U
$u = 0$	0.8772251458	0.9993155549
$u = 0.8t$	0.9140272726	0.9957985079
$u = 4t$	1.1237629942	1.0029796534
$u = -0.8t$	0.8453990280	1.0021154561
$u = 0.4 \sin(2t)$	0.9127125337	0.9958813085

The $u = 4t$ state initially grows, although its final norm in the table is below its initial value. For $u = 0$, the integrated loss is approximately 0.260973048240009. The largest recorded floating equality residual is below 6.7×10^{-14} ; it is diagnostic agreement, not an interval error bound or a proof of a continuum field theory.

The arithmetic-source program has 60 cases, four rational and 56 floating. It uses mpmath 1.3.0 at 40 decimal digits for the main source identities, and double-precision time integration with 25-digit function evaluations for the flow. Checks cover the xi logarithmic derivative, the complete prime/gamma source, the shift and Cayley equations, normalized source transport, finite model measures and the two obstruction controls. No xi-zero table is used; the finite polynomial models explicitly specify their own zeros.

For prime powers through $N = 1000$, the tail comparison uses

$$\sum_{n>N} \Lambda(n)n^{-\sigma} \leq N^{1-\sigma} \left(\frac{\log N}{\sigma-1} + \frac{1}{(\sigma-1)^2} \right), \quad (\text{A.1})$$

where $\sigma > 1$ and $(\log x)x^{-\sigma}$ is decreasing for $x \geq N$. The inequality follows from $\Lambda(n) \leq \log n$ and integral comparison. At $p = 5.5 + 0.7i$, $\omega = 0.2$, the floating residual is approximately 8.81×10^{-16}

against the analytical tail bound 6.24×10^{-15} . Evaluating a proven bound in floating arithmetic does not make the residual comparison an interval certificate.

For (6.14), at $\eta = 1/2$ and $\tau = 0.3$, the initial point $1 + 0.7i$ reaches approximately $1.2393712363 + 0.6045293214i$. The directly evaluated xi ratio is 1.0209360162 , agreeing with (6.15). Across the two sample trajectories the 64-versus-128-step difference is at most 1.36×10^{-13} and the xi linearization residual at most 7.31×10^{-16} . The composition check uses matching steps and primarily tests consistency; the separate xi identity independently tests the vector field.

Analytical arguments, rather than sign samples, establish the finite tensor balance, compact initial limit and safe positive-real source. The numerical work establishes neither the physical sewing operator nor unweighted arithmetic contraction or RH.

A.2 Source history and reproducibility

The main internal inputs are the 22 September notes *SU(2) level 2: a completed deterministic boundary Loewner pilot* and *The arithmetic source and an actual generalized Loewner evolution*, together with their same-assistant audits. The former followed the Chern–Simons/WZW proposal; the latter refined the priority toward arithmetic-source matching. The earlier commuting shift-flow and cumulative Cayley observations are credited in those notes, and are not new claims of this synthesis. The parent manuscript supplies the established causal operator conventions used in (7.8).

The local README links the preserved notes, both programs, their JSON records and replay instructions. The manuscript builds entirely from TeX files in this directory; it does not load parent manuscript sections. The build record identifies the delivered PDF, source files and unchanged research inputs. Such hashes establish file identity, not mathematical correctness. Future working milestones are indexed by commits or tags when available, without new dated manuscript snapshots.

A.3 Acknowledgements and review status

This manuscript was drafted for Edward Baker with substantial LLM assistance from GPT-6 (Codex, according to the developer-provided model identity), including mathematical synthesis, algebraic calculations, numerical programs, source checking and document preparation. The reasoning-effort setting was not exposed and is not inferred. Internal audits were performed by the same assistant; no independent specialist review is claimed. The classical results are attributed in the references, and unresolved mathematical and physical obligations are stated in Sections 7–8.

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